Physical AI for Science Hack · Track B (Opentrons)

Athelas Apothecary



From a plate design in the browser to a pipetting run on the Opentrons Flex

Towards fully automated, high-throughput drug screens with humans at the helm, on affordable hardware like the Opentrons Flex.

Team · You Shall Not Pipette (by hand)

Dose-response plates are still pipetted by hand

Slow

Every dilution series means planning volumes, intermediates and well positions on paper or in Excel.

Error-prone

One wrong C1·V1 = C2·V2 or a swapped column silently ruins a whole plate.

Gap to the robot

Liquid handlers are powerful, but going from an experiment idea to a safe protocol takes coding and trial runs.

Drug screens often require either error-prone pipetting by hand, because of costs, or expensive hardware. We change that.

One pipeline from order to robot

1

Design

Pick troughs, plate column and recipe in the web console

2

Plan & check

Volumes computed, plan replayed and checked well by well

3

Generate

Self-contained Flex protocol (API 2.30)

4

Simulate

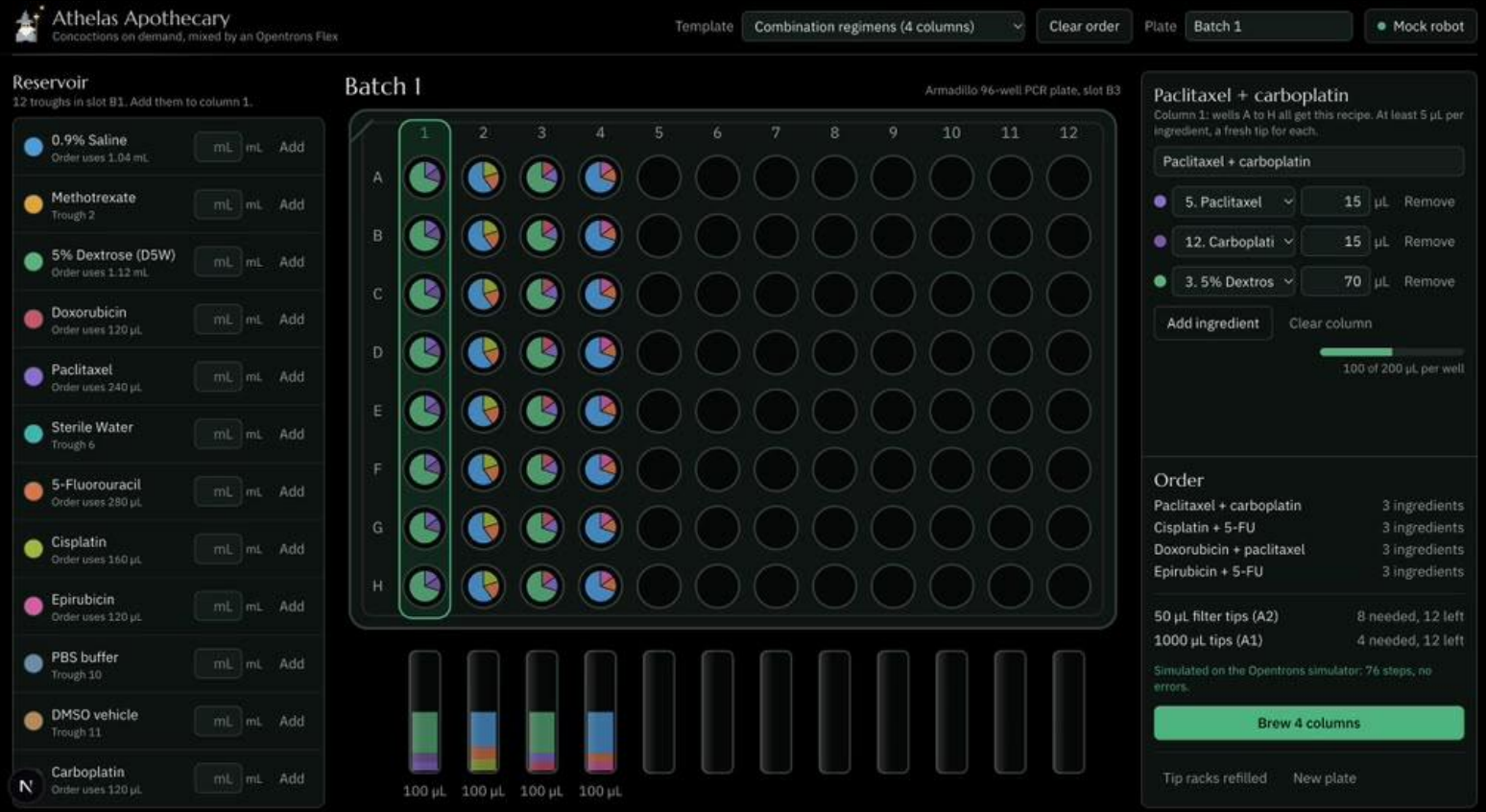
Opentrons 10.1.0 simulator gates every run

5

Brew

Run on the Flex or the mock robot, track troughs and tips

The console


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Concoctions on demand, mixed by an Opentrons Flex
Template
Combination regimens (4 columns)
Clear order
Plate
Batch 1
Mock robot
Reservoir
12 troughs in slot B1. Add them to column 1.
0.9% Saline
Order uses 1.04 mL
Methotrexate
Trough 2
5% Dextrose (D5W)
Order uses 1.12 mL
Doxorubicin
Order uses 120 µL
Paclitaxel
Order uses 240 µL
Sterile Water
Trough 6
5-Fluorouracil
Order uses 280 µL
Cisplatin
Order uses 160 µL
Epirubicin
Order uses 120 µL
PBS buffer
Trough 10
DMSO vehicle
Trough 11
Carboplatin
Order uses 120 µL
mL
Add
Batch 1
Armadillo 96-well PCR plate, slot B3
1 2 3 4 5 6 7 8 9 10 11 12
A B C D E F G H
100 µL 100 µL 100 µL 100 µL
Paclitaxel + carboplatin
Column 1: wells A to H all get this recipe. At least 5 µL per ingredient, a fresh tip for each.
Paclitaxel + carboplatin
5. Paclitaxel 15 µL Remove
12. Carboplati 15 µL Remove
3. 5% Dextros 70 µL Remove
Add ingredient
Clear column
100 of 200 µL per well
Order
Paclitaxel + carboplatin 3 ingredients
Cisplatin + 5-FU 3 ingredients
Doxorubicin + paclitaxel 3 ingredients
Epirubicin + 5-FU 3 ingredients
50 µL filter tips (A2) 8 needed, 12 left
1000 µL tips (A1) 4 needed, 12 left
Simulated on the Opentrons simulator: 76 steps, no errors.
Brew 4 columns
Tip racks refilled
New plate

Demo



Studio45 Flex deck



A1 1000 µL tips	**A2** 50 µL filter tips	**A3** Trash
B1 12-trough reservoir	**B2** 96 deep-well	**B3** PCR plate on riser
C1 free	**C2** free	**C3** free
D1 free	**D2** free	**D3** free

8-channel 1000 µL, left mount · gripper + deck camera · jog calibration before every real run

What went wrong on the first run



A little whoopsie daisy

Robot piercing pipette tips through a PCR plate. My mistake: I didn’t tell the robot that the PCR plate was on top of another plate so that it couldn’t wiggle. Now every fill refuses to run until jog calibration has measured the real plate height.

Built to be safe on a real robot

Simulation gate

Every order runs through the official Opentrons simulator before Brew is possible.

Independent plan checker

Replays the transfer list and checks every well; tested on 300 random layouts.

Pipette limits enforced

5 µL minimum, tip size chosen per volume, fresh tips per draw.

Mock robot

Rehearse the whole flow without hardware; API reachable only from the laptop.

Next steps

1. **384-well cell plates**
   CellCarrier-384 with a custom labware definition; 8-channel hits every other row (48 positions)

2. **Live well ledger**
   Every well’s contents, history and stock inventory, fed from the robot’s run log

3. **Barcodes & sample registry**
   Scan the stock plate, load its compounds, block runs on the wrong plate

Code, docs and data: github.com/LamaSu/Hardwarehack

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Code, docs & data